

Lecture 3 — Properties of OLS and Statistical Inference

Standard Errors, Significance, OLS Assumptions, and Causality (SW Chapters 4–7)

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Outline

- 1 From OLS algebra to statistical inference
- 2 Standard errors and statistical significance
- 3 Testing several restrictions: the F-test
- 4 Causal interpretation and OLS assumptions
- 5 OLS properties under the assumptions
- 6 Summary

From algebra to statistical inference

- Last lecture, OLS was an **algorithm**: choose the line that minimizes the sum of squared residuals.
- Today we first ask a mechanical statistical question: if we drew a different random sample, how much would $\hat{\beta}$ change?
- This leads to three practical objects in every regression table:

standard error $\implies$ t -statistic $\implies$ p -value / confidence interval.

- The **causal interpretation** of a coefficient is a separate question. We return to it after learning how to read uncertainty in a regression table.

First: precision and statistical significance. Later: when a coefficient can be read causally.

The variance of the OLS estimator

Assume the errors are **homoskedastic**: $\text{Var}(u_i | X) = \sigma^2$ for all i . Starting from $\hat{b} - b = T/S_{xx}$ and using $\text{Var}(a + bX) = b^2\text{Var}(X)$ with independence across i :

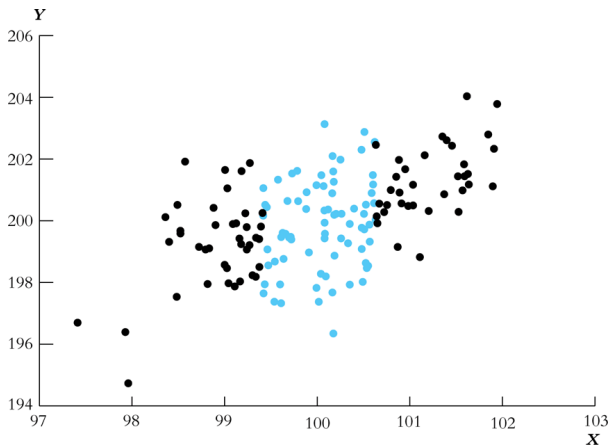
$$\text{Var}(\hat{b} | X) = \text{Var}\left(\frac{\sum_i (x_i - \bar{x})u_i}{S_{xx}}\right) = \frac{\sum_i (x_i - \bar{x})^2 \sigma^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}} = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}.$$

In matrix form:

$$\text{Var}(\hat{\beta} | X) = (X'X)^{-1}X'\sigma^2IX(X'X)^{-1} = \sigma^2(X'X)^{-1}.$$

- The variance *falls* with n (more terms in S_{xx}), *falls* with the spread of X , and *rises* with the noise σ^2 .

Why more variation in X helps



Source: Stock & Watson, *Introduction to Econometrics*, 4th ed. (Global Edition) — Figure 4.6

The math.

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2} \propto \frac{1}{n \text{Var}(X)}.$$

The intuition. If there is more variation in X , there is more information in the data with which to fit the line.

The number of black and blue dots is the same. Using which would you get a more accurate regression line?

The black points span a much wider range of X ; a line fitted to them is pinned down at both ends. The blue points are all bunched near $X = 100$ — small changes in the cloud swing the slope a lot.

The sampling distribution of $\hat{\beta}_1$

$\hat{\beta}_1$ is a random variable: a different sample of 420 districts gives a different number. Its exact distribution is complicated — it depends on the population distribution of (Y, X) — but for large n we have simple and good approximations.

If the three least squares assumptions hold, then as $n \rightarrow \infty$:

- $\hat{\beta}_1$ is **unbiased**: $\mathbb{E}[\hat{\beta}_1] = \beta_1$;
- $\hat{\beta}_1$ is **consistent**: $\hat{\beta}_1 \xrightarrow{p} \beta_1$ (law of large numbers);
- $\hat{\beta}_1$ is approximately **normally distributed** (central limit theorem):

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1)}} \xrightarrow{d} N(0, 1), \quad \text{Var}(\hat{\beta}_1) = \frac{1}{n} \frac{\text{Var}[(X_i - \mu_X)u_i]}{[\text{Var}(X_i)]^2}.$$

Exactly the same three results as for $\bar{Y}$ in Lecture 1 — with $\hat{\beta}_1$ in place of $\bar{Y}$. Everything we built there now applies here.

Degrees of freedom: why we divide by $n - k$

The sample-mean case. For i.i.d. $X_1, \dots, X_n$ with variance σ^2 ,

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

and taking expectations, using $\mathbb{E}[n(\bar{X} - \mu)^2] = n\text{Var}(\bar{X}) = \sigma^2$,

$$\mathbb{E}\left[\sum_i (X_i - \bar{X})^2\right] = n\sigma^2 - \sigma^2 = (n-1)\sigma^2 \quad \implies \quad \mathbb{E}\left[\frac{1}{n-1} \sum_i (X_i - \bar{X})^2\right] = \sigma^2.$$

The naive version $v_n = \frac{1}{n} \sum_i (X_i - \bar{X})^2$ is biased: $\mathbb{E}[v_n] = \frac{n-1}{n}\sigma^2$ (it is the MLE under normality, and consistent).

Rule of thumb: subtract the number of estimated parameters from n .

A concrete example of a lost degree of freedom

- Suppose we have three numbers: $x_1 = 2$, $x_2 = 4$, $x_3 = 6$.
- Sample mean: $\bar{x} = (2 + 4 + 6)/3 = 4$.
- Deviations from the mean: -2 , 0 , $+2$.
- Note $\sum_{i=1}^n (x_i - \bar{x}) = 0$ *always* holds. If you already know the first $n - 1$ deviations, the last one is completely determined — it must make the sum equal zero.
- That is why you lose one degree of freedom: with n data points you have only $n - 1$ free pieces of information about the variance.
- In regression with k estimated coefficients, the residuals satisfy k such constraints (the k normal equations), so you lose k degrees of freedom.

Estimating σ^2 in the regression

Model $y_i = \alpha + \beta x_i + u_i$, residuals $\hat{u}_i = y_i - \hat{\alpha} - \hat{\beta}x_i$, $\text{SSR} = \sum_i \hat{u}_i^2$. From the OLS algebra,

$$\hat{\alpha} - \alpha = \bar{u} - \bar{x} \frac{T}{S_{xx}}, \quad \hat{\beta} - \beta = \frac{T}{S_{xx}}, \quad \text{hence} \quad \hat{u}_i = (u_i - \bar{u}) - \frac{T}{S_{xx}}(x_i - \bar{x}).$$

Squaring and summing (using $\sum_i (x_i - \bar{x}) = 0$, $\sum_i (u_i - \bar{u})(x_i - \bar{x}) = T$, $\sum_i (x_i - \bar{x})^2 = S_{xx}$) gives the purely algebraic identity

$$\text{RSS} = \sum_{i=1}^n u_i^2 - n\bar{u}^2 - \frac{T^2}{S_{xx}}$$

- No distributional assumptions were used — only the definitions of T and S_{xx} .

Taking expectations $\Rightarrow$ an unbiased estimator

Using homoskedasticity and no autocorrelation:

$$\mathbb{E}\left[\sum_i u_i^2\right] = n\sigma^2, \quad \mathbb{E}[n\bar{u}^2] = \frac{\mathbb{E}[(\sum_i u_i)^2]}{n} = \sigma^2,$$

$$\mathbb{E}[T^2] = \text{Var}\left(\sum_i (x_i - \bar{x})u_i\right) = \sigma^2 \sum_i (x_i - \bar{x})^2 = \sigma^2 S_{xx} \implies \mathbb{E}\left[\frac{T^2}{S_{xx}}\right] = \sigma^2.$$

Therefore $\mathbb{E}[\text{SSR}] = n\sigma^2 - \sigma^2 - \sigma^2 = (n-2)\sigma^2$, and

$$\hat{\sigma}^2 = \frac{\text{SSR}}{n-2} = \frac{\sum_i \hat{u}_i^2}{n-2} \quad \text{is unbiased;} \quad \text{SER} = \hat{\sigma}.$$

With k coefficients the same argument gives $\hat{\sigma}^2 = \text{SSR}/(n-k)$, and under normality $(n-k)\hat{\sigma}^2/\sigma^2 \sim \chi_{n-k}^2$.

Recap: the testing framework

We built all of this for the sample mean; it transfers verbatim.

- A hypothesis test is a null $H_0 : \theta \in \Omega_0$, an alternative $H_1 : \theta \in \Omega_1$, and a decision rule: *reject H_0 if $W > c$* , where W is a statistic and c the critical value.
- A **type I error** rejects a true H_0 (*a bad grade for a good paper*); a **type II error** fails to reject a false one (*a good grade for a bad paper*).
- The **size** (significance level) is $\alpha = \max_{\theta \in \Omega_0} P_{\theta}\{W > c\}$; the critical value is $c_{\alpha} = F^{-1}(1 - \alpha)$ where F is the CDF of W under H_0 . The **power** is the probability of rejecting for $\theta \in \Omega_1$; a test is optimal if it minimizes type II error for a given size.
- The **p -value** is the smallest α at which you would reject: $p = 1 - F(W)$. **It is not the probability that H_0 is true** — H_0 is not random. It is a probability about the data, computed *assuming H_0* .

Three distributions we need

- **Chi-square.** If $Z_1, \dots, Z_m$ are independent $N(0, 1)$, then $Q = Z_1^2 + \dots + Z_m^2 \sim \chi^2(m)$, with mean m and variance $2m$.
- **Student t .** If $Z \sim N(0, 1)$ and $Q \sim \chi^2(m)$ are independent, then

$$T = \frac{Z}{\sqrt{Q/m}} \sim t(m).$$

Symmetric and bell-shaped like the normal, but with heavier tails; for m large it is indistinguishable from $N(0, 1)$.

- **Fisher F .** If $Q_1 \sim \chi^2(m_1)$ and $Q_2 \sim \chi^2(m_2)$ are independent, then

$$F = \frac{Q_1/m_1}{Q_2/m_2} \sim F(m_1, m_2).$$

If $T \sim t(m)$ then $T^2 \sim F(1, m)$; and for m_2 large, $m_1 F(m_1, m_2) \approx \chi^2(m_1)$.

The t -test for a single coefficient

Consider $y_i = x_{1i}\beta_1 + \cdots + x_{Ki}\beta_K + u_i$ with homoskedastic, independent, normally distributed errors. Because OLS is linear in y (hence in u), $\hat{\beta}$ is *normally distributed*.

Let $\widehat{\text{se}}(\hat{\beta}_k)$ be the standard error of $\hat{\beta}_k$ — the estimated standard deviation of its sampling distribution. Define

$$\hat{T} = \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\text{se}}(\hat{\beta}_k)}, \quad \text{typically with } \beta_{k,0} = 0 : \quad \hat{T} = \frac{\hat{\beta}_k}{\widehat{\text{se}}(\hat{\beta}_k)}.$$

Under $H_0 : \beta_k = \beta_{k,0}$, $\hat{T} \sim t(N - K)$ (and $\rightarrow N(0, 1)$ for large N).

- $N - K$: one degree of freedom lost for each estimated coefficient — the same bookkeeping as $\hat{\sigma}^2 = \text{SSR}/(N - K)$.

Two-sided test and confidence interval

Test $H_0 : \beta_k = 0$ against $H_1 : \beta_k \neq 0$. With F the CDF of $t(N - K)$,

$$\Pr\{|\hat{T}| > c\} = F(-c) + 1 - F(c) = 2[1 - F(c)],$$

so given α , choose $c_\alpha = F^{-1}(1 - \alpha/2)$, the $(1 - \alpha/2)$ -quantile. For $\alpha = 5\%$ and large N , $c_\alpha \approx 1.96$: **reject H_0 if $|\hat{T}| > 1.96$.**

p -value: $p = 2[1 - F(|\hat{T}|)]$. Reject at 5% if $p < 0.05$.

Confidence interval: as always, the set of null values not rejected,

$$\hat{\beta}_k \pm 1.96 \times \widehat{\text{se}}(\hat{\beta}_k).$$

Testing multiple hypotheses

Suppose we want to test the null

$$H_0 : \beta_1 = \beta_2 \quad \text{and} \quad \beta_3 + \beta_1 = 1,$$

against a general alternative with no constraints on $(\beta_1, \beta_2, \beta_3)$.

Two t -tests will not do: they do not control the joint error rate, and they cannot handle restrictions that link coefficients. The **Fisher (F) test** does.

Estimating the constrained model

Estimate $y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i$ ($K = 4$ including the intercept) both under H_0 and under H_1 .

Under the null, $\beta_2 = \beta_1$ and $\beta_3 = 1 - \beta_1$, so

$$y_i = \alpha + \beta_1 x_{1i} + \beta_1 x_{2i} + (1 - \beta_1) x_{3i} + u_i,$$

that is,

$$y_i - x_{3i} = \alpha + \beta_1 (x_{1i} + x_{2i} - x_{3i}) + u_i.$$

Regressing $y_i - x_{3i}$ on $(x_{1i} + x_{2i} - x_{3i})$ with an intercept gives the constrained OLS estimates of α and β_1 ; β_2 and β_3 then follow from the restrictions. Under H_1 , estimate the unconstrained model.

The general principle: imposing q restrictions turns a K -parameter regression into a $(K - q)$ -parameter one, and can only make the fit worse. The question is whether it gets *much* worse.

The Fisher test

Let SSR_0 be the sum of squared residuals of the *constrained* model and SSR that of the *unconstrained* model, with q restrictions. Define

$$\hat{F} = \frac{(SSR_0 - SSR)/q}{SSR/(N - K)}.$$

Under H_0 , $\hat{F} \sim F(q, N - K)$. Reject H_0 at level α if $\hat{F} > c_\alpha$, the $(1 - \alpha)$ quantile of $F(q, N - K)$.

- The numerator is the deterioration in fit caused by the restrictions, per restriction; the denominator is the noise level. $\hat{F}$ is a signal-to-noise ratio.
- $\hat{F} \geq 0$ always, so the F -test is one-sided by construction.
- With $q = 1$, $\hat{F} = \hat{T}^2$: the F -test and the t -test agree.

F -test example: “testing the model”

Test that all parameters but the intercept are zero. Under this null the only nonzero parameter is the constant, whose OLS estimator is $\bar{y}$ — so $SSR_0 = TSS$.

Therefore $SSR_0 - SSR = ESS = R^2 \times TSS$ and $SSR/TSS = 1 - R^2$, so with $q = K - 1$ restrictions

$$\hat{F} = \frac{R^2}{1 - R^2} \cdot \frac{N - K}{K - 1}$$

- This is the $F(1, 418) = 19.26$, $\text{Prob} > F = 0.0000$ line at the top of the STATA output. With one regressor, $19.26 \approx (-4.39)^2$ — the same test.
- Note $R^2 = 0.051$ is small and $\hat{F}$ is still overwhelming. Fit and significance are different things: with $n = 420$, even a weak relationship is precisely measured.

Reading the STATA output, at last

```
. regress testscr str, robust
```

Number of obs = 420

```
-----+-----  
      | Robust  
testscr| Coef. Std. Err. t P>|t| [95% Conf. Interval]  
-----+-----  
      str | -2.279808 .5194892 -4.39 0.000 -3.300945 -1.258671  
      _cons | 698.933 10.36436 67.44 0.000 678.5602 719.3057  
-----+-----
```

- $\hat{\beta}_1 = -2.28$ with $\hat{se} = 0.52$, so $\hat{T} = -2.28/0.52 = -4.39$. Since $|\hat{T}| > 1.96$, reject $H_0 : \beta_1 = 0$ at 5% (indeed $p < 0.001$).
- 95% CI: $-2.28 \pm 1.96 \times 0.52 = (-3.30, -1.26)$ — it excludes zero, which is the same statement as the test rejecting.
- `robust` asks for heteroskedasticity-robust standard errors, which do not require $\text{Var}(u | X) = \sigma^2$. Use them by default.

Causal interpretation: association vs. causal effect

- An association asks: do units with different X have different average Y ?

$$\mathbb{E}[Y \mid X = 20] - \mathbb{E}[Y \mid X = 25].$$

- A causal question asks: what would happen to the **same unit** if we changed X ?

$$Y_i(X = 20) - Y_i(X = 25).$$

- Example: smaller classes may have higher test scores because class size matters, or because richer districts have both smaller classes and better resources.
- Random assignment solves this by making X unrelated to other determinants of Y , so $\mathbb{E}[u \mid X] = 0$ becomes plausible.

OLS estimates an association mechanically. It estimates a causal effect only under additional assumptions about how X is generated.

OLS assumptions: what are they for?

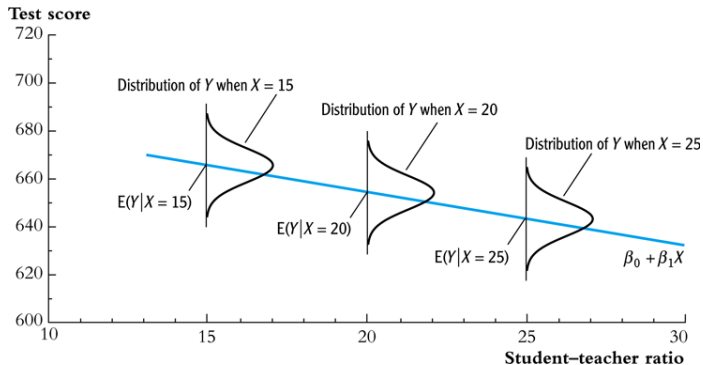
- Stock and Watson call these the **least squares assumptions**. Many texts simply call them **OLS assumptions** or **regression assumptions**.
- They are used for two related but distinct purposes:
 - ① **Statistical properties:** unbiasedness, variance formulas, standard errors, and tests.
 - ② **Causal interpretation:** if X is as-if randomly assigned, or if controls remove all confounding, then the slope can be read as a causal effect.

For the simple model $Y_i = \beta_0 + \beta_1 X_i + u_i$:

- ① **Exogeneity:** $\mathbb{E}[u \mid X = x] = 0$ for all x .
- ② **Random sampling:** (X_i, Y_i) are i.i.d.
- ③ **No large outliers:** extreme values of X and Y are rare.

For causality, exogeneity is the key: the omitted factors in u must not be systematically related to X .

LSA #1: $\mathbb{E}[u \mid X = x] = 0$ in a picture



Source: Stock & Watson, *Introduction to Econometrics*, 4th ed. (Global Edition) — Figure 4.4

- At each value of X there is a whole conditional distribution of Y .
- The assumption says the *mean* of each of these distributions lies exactly on the line: $\mathbb{E}[Y \mid X] = \beta_0 + \beta_1 X$.
- So the population regression line *is* the conditional expectation function — the best predictor of Y given X from Lecture 1.
- Nothing is assumed about the shape or the spread of those conditional distributions.

What exogeneity really requires

Example. $TestScore_i = \beta_0 + \beta_1 STR_i + u_i$, where u_i contains other determinants of test scores.

- If districts with smaller classes are richer, then family income is inside u_i and is correlated with STR_i .
- Then $\mathbb{E}[u \mid X = x] \neq 0$, and the slope mixes two things:
 - ① the effect of class size, and
 - ② differences in family income, school resources, teacher quality, etc.
- In an ideal randomized experiment, class size is assigned randomly. Then the other determinants in u_i are balanced across class sizes, so $\mathbb{E}[u \mid X = x] = 0$ is plausible.

The causal claim is not produced by OLS itself; it comes from the design or identifying assumption.

Normal equations vs. exogeneity

The same phrase “orthogonality” appears in three different places. They are related, but not the same.

Level	Condition	Meaning
Sample OLS algebra	$X'\hat{u} = 0$	Always true after OLS. The residuals are mechanically orthogonal to the regressors.
Population moment	$\mathbb{E}[Xu] = 0$	The population error is uncorrelated with the regressors.
Conditional exogeneity	$\mathbb{E}[u X] = 0$	Stronger: at every value of X , the omitted factors have mean zero.

$$\mathbb{E}[u | X] = 0 \implies \mathbb{E}[Xu] = 0, \quad X'\hat{u} = 0 \text{ does not prove either one.}$$

OLS always makes sample residuals orthogonal to X . Causal interpretation requires an assumption about the population error, especially $\mathbb{E}[u | X] = 0$.

Two consequences of exogeneity

Law of Iterated Expectations: $\mathbb{E}[Y] = \mathbb{E}(\mathbb{E}[Y \mid X])$ for any X, Y .

(i) The error has unconditional mean zero.

$$\mathbb{E}[u] = \mathbb{E}(\mathbb{E}[u \mid X]) = \mathbb{E}[0] = 0.$$

(ii) The error is uncorrelated with the regressor. Since x is a component of X ,

$$\mathbb{E}[xu] = \mathbb{E}(\mathbb{E}[xu \mid X]) = \mathbb{E}(x \mathbb{E}[u \mid X]) = \mathbb{E}[x \cdot 0] = 0,$$

and therefore $\text{Cov}(x, u) = \mathbb{E}[xu] - \mathbb{E}[x]\mathbb{E}[u] = 0$.

So exogeneity (“mean independence”) is *stronger* than zero correlation: it implies it, but not conversely. Zero correlation is all OLS algebra needs; mean independence is what a causal reading needs.

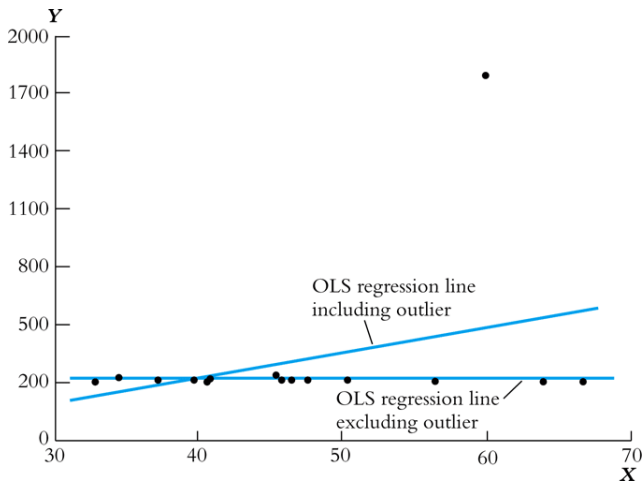
#2: (X_i, Y_i) **are i.i.d.** This arises automatically under simple random sampling:

- entities are selected from the same population, so (X_i, Y_i) are identically distributed;
- entities are selected at random, so different entities are independent.
- The main place we meet non-i.i.d. sampling is data recorded over time for the same entity (panel and time series data) — later in the course.

#3: large outliers are rare. A large outlier is an extreme value of X or Y .

- If X and Y are bounded they have finite fourth moments. (Standardized test scores automatically satisfy this; so do the STR, family income, etc.)
- The substance of the assumption is that a single large outlier can strongly influence the results — so we rule them out.
- **Look at your data!** If you have a large outlier: is it a typo? Does it belong in your data set? Why is it an outlier?

OLS can be very sensitive to one influential observation



- One unusual observation can substantially change the fitted regression line. Including it gives a strongly positive slope; excluding it gives an almost flat relationship.
- Why does this observation matter so much? It combines a very unusual Y value with an X value toward the edge of the sample. Together, they can pull the fitted slope upward.
- A large residual means that Y is far from what the line predicts. An unusual X value gives the observation more room to rotate the line.
- **Plot your data.** A regression table alone may not reveal which observations are driving the fitted line; a scatterplot often will.

Source: Stock & Watson, *Introduction to Econometrics*, 4th ed. (Global Edition) — Figure 4.5

What properties do we want from an estimator?

Before proving anything about OLS, it helps to separate three desirable properties.

- 1 **Correct target:** is the estimator centered on the parameter we want?

$$\mathbb{E}[\hat{\beta} \mid X] = \beta \quad (\text{unbiasedness})$$

- 2 **Precision:** how much does the estimator vary across repeated samples?

$$\text{Var}(\hat{\beta} \mid X) \quad \text{or its estimate, } se(\hat{\beta})$$

- 3 **Efficiency:** among comparable unbiased estimators, does it have the smallest variance?

“best” in BLUE = lowest variance within a class of estimators.

Unbiasedness asks whether we are centered correctly; standard errors ask how noisy the estimate is; BLUE asks whether OLS is the most precise linear unbiased estimator.

Bias, variance, and mean squared error

For an estimator $\hat{\beta}$ of β , a useful decomposition is

$$\underbrace{\mathbb{E}[(\hat{\beta} - \beta)^2 \mid X]}_{\text{mean squared error}} = \underbrace{\text{Var}(\hat{\beta} \mid X)}_{\text{variance}} + \underbrace{(\mathbb{E}[\hat{\beta} \mid X] - \beta)^2}_{\text{bias}^2}.$$

- **Unbiasedness:** the second term is zero.
- **Small standard error:** the first term is small.
- **Gauss–Markov:** within linear unbiased estimators and homoskedastic errors, OLS has the smallest variance.

Important limitation: “best linear unbiased” does not mean best among all possible estimators. A biased estimator can sometimes have lower mean squared error.

Roadmap: what we will prove for OLS

- **Unbiasedness:**

$$\hat{\beta} = \beta + (X'X)^{-1}X'u \quad \Rightarrow \quad \mathbb{E}[\hat{\beta} \mid X] = \beta \quad \text{if } \mathbb{E}[u \mid X] = 0.$$

- **Variance:**

$$\text{Var}(\hat{\beta} \mid X) = \sigma^2(X'X)^{-1} \quad \text{under homoskedastic errors.}$$

- **BLUE:** if an estimator is *linear in y* and *unbiased*, its variance cannot be smaller than the OLS variance.

The logic is: assumptions $\rightarrow$ unbiasedness and variance formula $\rightarrow$ standard errors and statistical inference.

The OLS estimator is unbiased

Model $y_i = a + bx_i + u_i$. Since $y_i - \bar{y} = b(x_i - \bar{x}) + (u_i - \bar{u})$ and $\sum_i (x_i - \bar{x}) = 0$,

$$\hat{b} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = b + \frac{\sum_i (x_i - \bar{x})u_i}{\sum_i (x_i - \bar{x})^2}.$$

$$\hat{b} - b = \frac{\sum_i (x_i - \bar{x})u_i}{\sum_i (x_i - \bar{x})^2} = \frac{T}{S_{xx}}, \quad \text{where } T := \sum_i (x_i - \bar{x})u_i, \quad S_{xx} := \sum_i (x_i - \bar{x})^2.$$

Taking conditional expectations and using $\mathbb{E}[u_i \mid x_i] = 0$,

$$\mathbb{E}[\hat{b} \mid x_1, \dots, x_n] = b, \quad \text{hence} \quad \mathbb{E}[\hat{b}] = b,$$

and similarly $\mathbb{E}[\hat{a}] = a$.

This one-line decomposition, $\hat{b} = b + T/S_{xx}$, does all the work in the rest of the lecture: unbiasedness, the variance formula, and the sampling distribution all come straight out of it.

Unbiasedness vs. causality

- **Unbiasedness is a statistical property of an estimator.**

$$\mathbb{E}[\hat{\beta}_1] = \beta_1$$

It says that repeated-sample estimates are centered on the target parameter β_1 .

- **Causality is an interpretation of the target parameter.**

$$\beta_1 = \text{effect of changing } X \text{ on } Y$$

This requires an argument about how X is assigned or determined.

- OLS can be unbiased for a *non-causal association*. For a causal interpretation, X must be unrelated to the other determinants of Y contained in u .

$$\text{credible exogeneity } \mathbb{E}[u \mid X] = 0 \quad \Rightarrow \quad \begin{cases} \text{OLS is centered on } \beta, \\ \beta \text{ may be interpreted causally.} \end{cases}$$

Unbiasedness: are we estimating β correctly? Causality: is β the effect we care about?

When exogeneity fails, OLS is biased

Start from the same decomposition:

$$\hat{\beta} = \beta + (X'X)^{-1}X'u.$$

Taking conditional expectations given the regressors,

$$\mathbb{E}[\hat{\beta} \mid X] - \beta = (X'X)^{-1}X'\mathbb{E}[u \mid X].$$

- If $\mathbb{E}[u \mid X] = 0$, the bias term is zero.
- If $\mathbb{E}[u \mid X] \neq 0$, then in general

$$\mathbb{E}[\hat{\beta} \mid X] \neq \beta.$$

- If the regression error is systematically related to the regressors, OLS partly attributes the effect of the error component to X .

Exogeneity is exactly the assumption that removes the bias term. Without it, OLS can be precise but centered on the wrong target.

Example: omitted-variable bias

Suppose the true wage equation is

$$\log(wage_i) = \beta_0 + \beta_1 education_i + \beta_2 ability_i + u_i, \quad \beta_2 > 0.$$

But we estimate a regression that omits ability:

$$\log(wage_i) = \alpha_0 + \alpha_1 education_i + v_i, \quad v_i = \beta_2 ability_i + u_i.$$

If

$$\text{Cov}(education_i, ability_i) > 0,$$

then the new error term v_i is correlated with education, so exogeneity fails.

In the simple linear case,

$$\alpha_1 = \beta_1 + \beta_2 \frac{\text{Cov}(education_i, ability_i)}{\text{Var}(education_i)}.$$

- The short regression does not isolate the causal effect of education, β_1 .
- It also picks up part of the effect of ability.

Reading the omitted-variable-bias formula

Write the bias as

$$\alpha_1 - \beta_1 = \underbrace{\beta_2}_{\text{effect of omitted variable on } Y} \times \underbrace{\frac{\text{Cov}(\text{education}, \text{ability})}{\text{Var}(\text{education})}}_{\text{relationship with included } X}.$$

- If ability raises wages ($\beta_2 > 0$), and more educated workers tend to have higher ability, the education coefficient is biased upward.
- If the omitted variable is unrelated to education, there is no omitted-variable bias from that variable.
- If the omitted variable does not affect wages, there is no omitted-variable bias from that variable.

Takeaway: Bias requires two ingredients: the omitted variable affects Y , and it is correlated with the included regressor.

Unbiasedness: vector version

Substitute $y = X\beta + u$ into $\hat{\beta} = (X'X)^{-1}X'y$:

$$\hat{\beta} = (X'X)^{-1}X'(X\beta + u) = \beta + (X'X)^{-1}X'u.$$

Take conditional expectations given X :

$$\mathbb{E}[\hat{\beta} \mid X] = \beta + (X'X)^{-1}X' \mathbb{E}[u \mid X] = \beta,$$

and by iterated expectations $\mathbb{E}[\hat{\beta}] = \beta$.

What each assumption does here:

- The *rank condition* ($\text{rank}(X) = k$) makes $X'X$ invertible, so $\hat{\beta}$ is even well-defined.
- *Exogeneity* ($\mathbb{E}[u \mid X] = 0$) rules out omitted-variable bias and other forms of endogeneity. If $\mathbb{E}[u \mid X] \neq 0$, then in general $\mathbb{E}[\hat{\beta} \mid X] \neq \beta$ and OLS is **biased**.

Setup. $y = X\beta + u$ with fixed $X \in \mathbb{R}^{n \times k}$.

Definition. An estimator $\hat{\beta}$ is *linear (in y)* if $\hat{\beta} = A(X)y$, where $A(X)$ may depend on X but *not* on y .

- OLS: $A_{\text{OLS}}(X) = (X'X)^{-1}X'$.
- GLS/WLS with known weights: $A(X) = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}$.

Do not confuse two uses of “linear”.

- A *linear model* is linear **in the parameters**: $\mathbb{E}[y \mid X] = X\beta$. Non-linear terms like x^2 or $\log x$ keep the model linear in β .
- A *linear estimator* is linear **in the data** y .

What unbiasedness forces

Unbiasedness for all β requires

$$\mathbb{E}[\tilde{\beta} \mid X] = A(X) \mathbb{E}[y \mid X] = A(X)X\beta = \beta \quad \forall \beta \quad \Longrightarrow \quad \boxed{A(X)X = I_k.}$$

OLS satisfies it: $A_{\text{OLS}}(X)X = (X'X)^{-1}X'X = I_k$.

Every linear unbiased estimator is OLS plus something orthogonal to X . Define

$$C(X) := A(X) - (X'X)^{-1}X' \quad \Longrightarrow \quad C(X)X = 0,$$

so that

$$\tilde{\beta} = \hat{\beta}_{\text{OLS}} + C(X)y, \quad C(X)X = 0.$$

This decomposition is the whole of the Gauss–Markov proof.

The Gauss–Markov theorem: OLS is BLUE

- The variance of any linear unbiased $\tilde{\beta}$:

$$\text{Var}(\tilde{\beta} \mid X) = \text{Var}((X'X)^{-1}X'y + Cy) = \sigma^2 \left((X'X)^{-1} + CC' \right),$$

where the cross terms vanish because $CX = 0$.

- Therefore

$$\text{Var}(\tilde{\beta} \mid X) - \text{Var}(\hat{\beta}_{\text{OLS}} \mid X) = \sigma^2 CC' \succeq 0,$$

since $a'(\sigma^2 CC')a = \sigma^2 \|C'a\|^2 \geq 0$ for every vector a .

OLS has the smallest covariance matrix among all linear unbiased estimators: **OLS is the Best Linear Unbiased Estimator.**

- Note what it does *not* say: nothing about non-linear estimators, nothing if the errors are heteroskedastic, and nothing about bias–variance trade-offs (Ridge and Lasso are biased and can beat OLS in mean squared error).

Summary

- **Three least squares assumptions:** $\mathbb{E}[u \mid X] = 0$; i.i.d. sampling; no large outliers. Plus the rank condition so that $\hat{\beta}$ exists.
- Under them, $\hat{\beta}$ is **unbiased** and **consistent**, with $\text{Var}(\hat{\beta} \mid X) = \sigma^2(X'X)^{-1}$, and approximately **normal** in large samples.
- **Gauss–Markov:** with homoskedastic errors, OLS is BLUE.
- $\hat{\sigma}^2 = \text{SSR}/(n - k)$ is unbiased for σ^2 ; its square root is the SER.
- Inference is the same three-step recipe as for a mean: **estimate**, compute the **standard error**, form $t = (\hat{\beta}_k - \beta_{k,0})/\widehat{\text{se}}$; reject if $|t| > 1.96$, and report $\hat{\beta}_k \pm 1.96 \widehat{\text{se}}$.
- For several restrictions at once, use the F -test.

Only the formula for the standard error ever changes.

What is still unresolved

- We can now say that $\hat{\beta}_1 = -2.28$ is precisely estimated and very unlikely to be zero.
- We *cannot* yet say it is the causal effect of class size. That claim rests entirely on LSA #1, $\mathbb{E}[u \mid STR] = 0$ — and districts with small classes differ from districts with large ones in many other ways.
- **Next:** multiple regression as a tool against omitted variable bias; what happens to $\hat{\beta}_1$ when a relevant variable is left in u ; and how to choose control variables.